

MovementLogger — Sensor & Module Catalog

5 Teile · 2026-05-24 17:03 UTC · BOM aus movement_logger_firmware · via crawl2pump

5 neu (7 Tage)

0 aktualisiert

GPS / GNSS 10

SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK)

OSS firmware

Qwiic / I²C

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

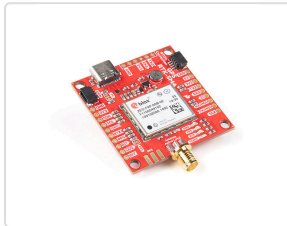
☐ Motion sensors

☐ SD-card

L×B×H 4.6 × 4.3 × 0.6 cm · 10 g

Datasheet: https://content.u-blox.com/sites/default/files/ZED-F9P-04B_DataSheet_UBX-21044850.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK) · GPS-16481 @ DigiKey

CHF 214.23

SparkFun @ DigiKey · in stock

DIY building block for a custom RTK tracker. u-blox ZED-F9P multi-band (L1+L2 GPS / Galileo / GLONASS / BeiDou) — same chip as the RTK Facet above, but as a bare Qwiic breakout for board-mount builds. Pair with any ESP32 dev board in this BOM (SparkFun Thing Plus-C, Seeed XIAO ESP32-S3) for WiFi + an NTRIP client; SparkFun ships the Arduino library `SparkFun u-blox GNSS Arduino Library` that wraps UBX protocol over I²C/Qwiic. SMA antenna jack onboard — wire to the u-blox ANN-MB-00 multi-band antenna below (L1+L2 active patch, 5 m coax, magnetic mount) for production performance. Without RTK corrections this is still a ~30 cm receiver under good sky; with RTK it's 1–2 cm. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/16481/12178437>

SparkFun RTK Facet L-Band (turnkey RTK tracker, ESP32 + ZED-F9P)

OSS firmware

USB-C pluggable

☒ USB-C

☒ WiFi

☒ Bluetooth

☒ GPS

☒ Motion sensors

☒ SD-card

L×B×H 13.4 × 9.3 × 2.9 cm · 245 g

MCU: ESP32 (WROOM-32E) · dual Xtensa LX6 @240 MHz · 520 KB SRAM · WiFi+BLE

Datasheet: <https://www.sparkfun.com/products/20000>

OSS firmware: https://github.com/sparkfun/SparkFun_RTK_Firmware



neu SparkFun RTK Facet L-Band (turnkey RTK tracker, ESP32 + ZED-F9P) · SparkFun 20000 @ sparkfun.com

USD 1099.95

SparkFun

Turnkey high-precision GNSS tracker. u-blox ZED-F9P multi-band (L1+L2) RTK receiver + ESP32-WROOM-32E (WiFi+BLE) + helical L-Band antenna + integrated LiPo + microSD logger + OLED + IMU bubble-level, in a single survey-grade unit. Three correction paths: (a) ****L-Band SSR**** via u-blox PointPerfect — ~10 cm accuracy, no internet, satellite-broadcast — needs a PointPerfect subscription (~USD 300/year) but works anywhere with sky view; (b) ****NTRIP over WiFi**** — ~1.4 cm accuracy, free in CH via swisstopo SWIPOS, needs WiFi or a phone tether to the spot; (c) ****base station / RTCM over BT**** — same accuracy as NTRIP, no internet, requires you to set up the base. The firmware (github.com/sparkfun/SparkFun_RTK_Firmware) handles all three out of the box. Logs raw + parsed NMEA to microSD; streams live position over WiFi or BLE to a phone (Bluetooth-SPP compatible with SW Maps, Field Genius, QGIS,...)

<https://www.sparkfun.com/products/20000>

u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 6 × 6 × 1.6 cm · 120 g

Datasheet: https://content.u-blox.com/sites/default/files/ANN-MB_DataSheet_UBX-18049862.pdf



neu **u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m) · ANN-MB-00 @ DigiKey**

CHF 48.07

u-blox @ DigiKey · in stock

Multi-band L1/L2/L5 active patch antenna for the ZED-F9P. Magnetic base, 5 m RG-174 coax, SMA male plug — the antenna u-blox themselves spec for multi-band testing. Without a multi-band antenna the ZED-F9P falls back to L1-only and you lose the RTK precision advantage; pair these two. Mount the puck on a non-metal ground-plane (the foilboard deck or case lid) with clear sky view. · SMA / U.FL · firmware: open-source

<https://www.digikey.com/en/products/detail/u-blox/ANN-MB-00/9817928>

WiFi Modules 5

Seeed Studio XIAO ESP32-S3 (WiFi + BLE, USB-C)

OSS firmware

USB-C pluggable

☒ USB-C

☒ WiFi

☒ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 2.1 × 1.75 × 0.4 cm · 3 g

MCU: ESP32-S3 · dual Xtensa LX7 @240 MHz · 8 MB flash / 512 KB SRAM (+PSRAM) · WiFi+BLE

OSS firmware: <https://github.com/espressif/esp-idf>



neu **Seeed Studio XIAO ESP32-S3 (WiFi + BLE, USB-C) · XIAO ESP32S3 @ DigiKey**

CHF 5.83

Seeed Studio @ DigiKey · in stock

Xtensa XIAO, USB-C. OSS ESP-IDF / Arduino. · USB-C · firmware: open-source

<https://www.digikey.com/en/products/detail/seeed-technology-co-ltd/113991114/19285530>

SparkFun Thing Plus - ESP32 WROOM (USB-C)

OSS firmware

USB-C pluggable

☒ USB-C

☒ WiFi

☒ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 5.84 × 2.29 × 0.71 cm · 8 g

MCU: ESP32 (WROOM) · dual Xtensa LX6 @240 MHz · 520 KB SRAM · WiFi+BLE

OSS firmware: <https://github.com/espressif/esp-idf>



neu **SparkFun Thing Plus - ESP32 WROOM (USB-C) · WRL-20168 @ DigiKey**

CHF 27.05

SparkFun @ DigiKey · out of stock

USB-C ESP32 + Qwiic. OSS Arduino/ESP-IDF, open hardware. · USB-C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/20168/16688100>